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Toxic effects of polyethylene microplastics on transcriptional changes, biochemical response, and oxidative stress in common carp (*Cyprinus carpio*)

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ABSTRACT

Aquatic ecosystems have become a place for accumulating microplastics (MPs). MPs can directly or indirectly damage organisms. Although studies of the toxicity of MPs, there are insufficient literature reports on the effects of MPs on freshwater aquatic life. Therefore, this study aimed to evaluate the effect of MPs toxicity on *Cyprinus carpio*. In this study, biochemical parameters, oxidative biomarkers, and gene expression were assayed in fish exposed to 0, 175, 350, 700, and 1400 μg L^{-1} of MPs for 30 days. MPs were detected in the liver and intestine of fish using FTIR-analysis. *Mt1*, *Ces2*, and *P450* mRNA expression were enhanced in the hepatocytes of fish exposed to MPs, while *Mt2* gene expression was significantly decreased. After exposure to MPs, MDA and carbonyl protein levels were higher than those of the reference group. The antioxidant capacity and glycogen contents in the hepatocytes significantly declined. MPs significantly inhibited glutathione reductase (GR), glucose 6-phosphate dehydrogenase (G6PDH), and catalase (CAT) activities. However, superoxide dismutase (SOD) and glutathione peroxidase (GPx) activities increased. MPs decreased the total protein, globulin levels, and butyrylcholinesterase (BChE) activity in blood. In contrast, aspartate aminotransferase (AST), alanine aminotransferase (ALT), gamma-glutamyl transferase (GGT), lactate dehydrogenase (LDH), alkaline phosphatase (ALP), and creatine phosphokinase (CPK) activities increased in treated-fish with MPs. Glucose, creatinine, cholesterol and triglyceride concentrations in fish exposed to MPs were significantly higher than that of the reference group. Consequently, MPs exposure could disrupt biochemical homeostasis, oxidative stress and alter the expression of genes involved in detoxification.

1. Introduction

In recent decades, the global production of plastics has increased significantly. In 2018, over 400 million tons of plastics were produced of which an estimated 13 million tons were in the form of plastic waste and debris left in aquatic ecosystems (Cunha et al., 2020). Similarly, if we assume a constant plastic production rate, its global production will double in 2025 and more than triple by 2050. Approximately 10 % of the plastics produced globally are found as plastic wastage in water ecosystems (Iheanacho et al., 2020; Zhang et al., 2021). Estimates show that >13–18 million tons of plastic waste and debris enter the world's oceans annually (Cunha et al., 2020; Zhang et al., 2021), which can increase the concentration of microplastics (MPs) to over 320 μg^{-1} (Zhang et al., 2021). In the study of mineral water samples in plastic bottles, the concentration of MPs was estimated to be 657 ± 633 μg^{-1} or $5.42 \times 10^7 \pm 1.95 \times 10^7$ particles per liter (Zuccarello et al., 2019).

Currently, plastic waste accounts for 80 % of the total waste in aquatic ecosystems (Cunha et al., 2020). Studies in Dutch waters show that the average concentration of MPs could reach >213,147 particles per cubic meter of water (Mughini-Gras et al., 2021). It is estimated that at least 5.25 trillion plastic particles float in seawater and oceans due to the volume of plastic waste discharged into waters (Frias et al., 2016; Qu et al., 2019). This estimate focuses solely on floating plastic particles and does not include the amount and volume of plastic submerged and deposited in the seabed and oceans (Dong et al., 2021). Frias et al. (2016) showed that 80 % of plastic waste originates from land. Moreover, >70 % of this waste accumulates in the seabed and oceans (Frias et al., 2016).

Plastic debris in the environment can break down into smaller pieces by physical erosion, the sun's ultraviolet rays, and physical or chemical

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processes (Dong et al., 2021). Plastic particles are classified into mega (>10 cm), macro (5–10 cm), *meso* (1–2.5 cm), micro (<5–1 mm) and nano (<0.5 mm) plastics (Parker et al., 2020). MPs are the most important plastics in aquatic ecosystems.

MPs are found in relatively high concentrations in marine and estuarine ecosystems and may accumulate in the bodies of freshwater and marine fish species (Strungaru et al., 2019).

Fish may mistakenly devour small plastics that are similar to their natural prey. Transmission of MPs along the food chain can occur through ingestion of MPs by aquatic organisms. MPs may be excreted in the faeces or enter the bloodstream after being absorbed through the intestinal epithelium and distributed to other tissues in the body (Kim et al., 2021). Micro and nanoparticles may penetrate the cellular membrane and enter the bodies of fishes. Once MPs are uptake in the digestive system, they can penetrate the anaerobic tissues of the intestine and accumulate in the adipose tissue of animals (Kim et al., 2021). The mode of exposure to MPs and the route of exposure (aquatic or nutritional) significantly affect the bioaccumulation of MPs. MPs can lead to oxidative damage, cytotoxicity, endocrine dysfunction, immunosuppression, neurotransmitter dysfunction, genetic toxicity and even fish death (Banaee et al., 2019c; Ding et al., 2020). Furthermore, the tiny pieces of plastic may obstruct respiratory and gastrointestinal systems.

Previous studies showed that exposure of aquatic animals to MPs can affect hematological and biochemical parameters (Kim et al., 2021; Banaee et al., 2021). Furthermore, bio-accumulation of MPs (Ding et al., 2018), gastrointestinal dysfunction (Weber et al., 2021), and histopathological damage (Hamed et al., 2021a, 2021b, 2021c), and genotoxicity (Sun et al., 2021) are reported in the fish exposed to MPs. Kim et al. (2021) declared that exposure to MPs could induce neurotoxicity and oxidative damage. A decreased appetite (Critchell and Hoogenboom, 2018), disorders of cellular metabolism (Watts et al., 2014), and an increased mortality rate (Oliveira et al., 2013) were also seen in the aquatic animals after exposure to MPs. Furthermore, they can play an important role in transmitting other pollutants like the Trojan horse (Banaee et al., 2019c; Strungaru et al., 2019; Xu et al., 2021). The most critical hypothesis in this study is that MPs can affect the physiology and health of fish. We used common carp (*Cyprinus carpio*) in the current study because *C. carpio* is an important commercial fish in Iran. This study aimed to evaluate the change in biochemical parameters, gene expression and oxidative stress of carp exposed to MPs.

2. Materials and methods

2.1. Animals

All the techniques, animal care and experimental procedure used in the present study followed the instructions and regulations set by Islamic Azad University, Marvdasht Branch, Iran. Juvenile common carp (*Cyprinus carpio*) used in the study were obtained from a local fish farm (Shiraz, Iran) and transferred to the laboratory. Then, fish of similar sizes (109.5 ± 15.5 g, 20.8 ± 1.2 cm) were randomly distributed into 15 fiberglass tanks (300 L) at a density of 15 fish per tank. Fish then were acclimated for experiment conditions (temperature: 24.5 ± 1.5 °C, pH: 7.4 ± 0.2; total hardness: 250 ± 5 mg L^{-1} $CaCO_3$; dissolved oxygen: 6.5 ± 0.5 mg L^{-1}; photoperiod: 14 light/10 dark) and fed on a commercial diet (Bayza Feed Mill, Iran) two weeks before the experiment. Water exchange was done daily.

2.2. Experimental design

Fish were divided into five groups and exposed to different concentrations of MPs (0, 175, 350, 700, 1400 μg L^{-1}) for 30 days. In the present study, High-density polyethylene (PE100) powder (<0.2 mm; Eshragh Trading Co., Iran) was used as MPs. Concentrations of MPs were chosen based on previous studies (Zhang et al., 2021). Each group was kept in three replicates, and each replicate contained 15 fish.

After 30 days, six fish per tank were caught randomly to get the blood sample from the caudal vein with a 2.5-mL syringe containing anticoagulant. The blood samples were centrifuged at 6000 rpm for 10 min at 4 °C (DOMEL). Next, supernatant (plasma) was separated and stored in a freezer at −80 °C until analysis. Then, fish were dissected to collect the liver samples. The liver was quickly removed and aliquot into two parts. The first part of the liver was placed in liquid nitrogen (−196 °C) for gene expression analysis, and the second amount was homogenized into a mixture solution containing 100 mM cold potassium phosphate buffer (Sigma-Aldrich, Germany) pH 7.0, with 2 mM EDTA (Riedel-Haën, Germany). Then, this mixture was centrifuged at 15,000*g*-force for 15 min at 4 °C, and the supernatants were gathered and kept up at −80 °C until further analysis.

2.3. Blood biochemical parameters

Activities of aspartate aminotransferase (AST) or glutamate oxalate transaminase (SGOT), and alanine aminotransferase (ALT) or glutamate pyruvate transaminase (SGPT) were measured using α-ketoglutarate as a substrate (Moss and Henderson, 1999). Lactate and NAD+ as a substrate was used to detect lactate dehydrogenase (LDH) activity (Moss and Henderson, 1999). In measuring gamma-glutamyl transferase (GGT) activity, L-gamma-glytamyl-3carboxy-4-nitroanilide and glycylglycine were used as a substrate (Moss and Henderson, 1999). Alkaline phosphatase (ALP) activity was assayed using *P*-Nitrophenyl phosphate as a substrate (Moss and Henderson, 1999). According to the creatine phosphokinase (CPK) activity assay instructions, creatine was used as a substrate (Moss and Henderson, 1999). In estimating butyrylcholinesterase (BChE) activity, butyrylcholine was applied as a substrate (Johnson and Moore, 2012). Glucose concentration was measured in the presence of glucose oxidase (Sacks, 1999). Total protein was estimated by Biuret's method (Johnson et al., 1999) using copper sulphate as a substrate. Cholesterol levels were evaluated using cholesterol esterase and cholesterol oxidase according to the method described by Rifai et al. (1999). Triglycerides levels were determined in the presence of lipoprotein lipase and glycerol kinase (Rifai et al., 1999). Creatinine content was calculated in the presence of NaOH and picric acid. All biochemical parameters were measured according to the instructions of the diagnostic biochemical kit (ParsAzemun Co, Iran) by a UV–Vis spectrophotometer (Unico 2100 UV–Visible).

2.4. Oxidative biomarkers

Glutathione peroxidase (GPx), glutathione reductase (GR), glucose 6-phosphate dehydrogenase (G6PDH), and Superoxide dismutase (SOD) activities in the hepatocytes were evaluated according to the biochemical kit instructions bought from the Biorexfars Co., Iran. Catalase (CAT) activity was estimated according to the instructions presented by Góth (1991) using a hydrogen peroxide solution (30 mM) as a substrate (Góth, 1991). The total antioxidant capacity was measured following the ferric reducing ability of plasma (FRAP) procedure using TPTZ (2,4,6-Tris(2-pyridyl)-*s*-triazine) as a substrate (Benzie and Strain, 1996). Malondialdehyde (MDA) as a lipid peroxidation metabolite in the hepatocytes was assayed using the modified method offered by Draper and Hadley (1990), utilizing thiobarbituric as substrate acid and monitored at 532 nm. Protein carbonyl was assayed with 2,4-dinitrophenylhydrazine (DNPH) at 370 nm according to the method provided by Colombo et al. (2016). Glycogen content in the liver was assayed by the protocol presented by Zhang (2012).

2.5. Gene expression

2.5.1. Total RNA isolation and reverse transcription

A piece of the common carp liver was dissected, dipped in TRIzol Reagent (RiboEx, South Korea), and put in a freezer (at −80 °C) until samples were used. According to the kit's protocol, to extract the total

RNA, TRIzol Reagent was added to 100 mg of homogenized hepatocytes. Then, the RNA samples were exposed to RNase-free water with DEPC (Thermo Scientific, USA). Next, to prevent genomic contamination, the samples were exposed to DNase I (Thermo Scientific, USA) for 30 min at 37 °C. Then, the quality and quantity of total RNA were examined by electrophoresis (Biorad, Minigel) and Nano-drop spectrophotometer (Thermo Scientific NanoDrop 2000, USA), respectively. According to the manufacturer's instructions, the cDNA was prepared using an oligo-dT primer and reverse transcriptase (Thermo Scientific), (Derikvandy et al., 2020).

2.5.2. Primer design

Primer design of 4 genes involved in detoxification processes (*P450*, *Ces2*, *Met1*, and *Met2*) was performed by OLIGO Software and performed in NCBI (Table 1).

2.5.3. Quantitative real-time PCR (RT-qPCR)

The amplification was performed with each primer to optimize the annealing temperature for the respective gene. The total volume was 25 μL, including 10 μL of SYBR Green Master Mix reagents (2×), 2 μL of cDNA and 0.8 μL of each primer (10 μM), ROX dye (0.4 μL) and 11 μL DEPC H2O. The qRT-PCR analysis was carried out using the StepOne-Plus Real-Time PCR system with the following protocols: initial denaturation at 94 °C for 5 min, followed with 40 cycles of denaturation at 94 °C for 30 s, annealing at 60 °C for 30 s and extension at 72 °C for 30 s. An extra temperature-ramping step from 95 °C to 65 °C was set for the melting curve. Relative gene expression was computed using the comparative CT (2ˆ−ΔCT) (Livak and Schmittgen, 2001).

2.6. FTIR analysis

After the autopsy, the intestines and liver of the fish were separated and placed in liquid nitrogen. The samples were then dried in a freezer dryer. Next, polyethylene (MPs) were detected in the tissues by FTIR Spectrometer (Bruker). FTIR could detect the high density of polyethylene in the 1445–1650 wavenumber (cm^{-1}) (Li et al., 2022a, 2022b).

2.7. Statistical analysis

First, the normality of the data was assayed using the Shapiro-Wilk test. Next, statistical analysis of all data were done using Graph Pad Prism 8 software. One-way analysis of variance (ANOVA) and Duncan test were used to compare the mean biochemical results and evaluate the significance of the differences between the experimental groups. Results are shown as mean ± S.D in each experimental group. It was considered a statistically significant difference when $P < 0.05$.

3. Results

Considering the light transmittance (%) was identified at wavenumbers 1445–1650 cm^{-1}, the MPs accumulation in the intestines and liver of fish could be confirmed by the Fourier transform infrared spectroscopy (FTIR) (Figs. 1 and 2; Table 2).

Table 1
Primer sequences, amplicon length, and temperature adjust instructions are used for annealing in Real-time PCR reaction.

Gene name	Primer	Sequence (5-3)	Tm	Length (bp)
Ces2	Forward	ACCCTCCATCACAGTTGCCTC	62	116
	Reverse	CGGCCTTCACAAAACTGGGTC	62	116
P450A	Forward	GACCTCTGCCTTTTGTGGGA	61	119
	Reverse	TATCGCTGGCTTTCTCCCGA	61	119
Mt1	Forward	CTGTTCTTGTTGCCCGTCTG	59	110
	Reverse	ACAAAACATCACGTTGACCTCC	59	110
Mt2	Forward	GAGGGACTTTCGGGCTCTTT	59	150
	Reverse	GGGCAACAGGAACAGCAACT	59	150
β-actin	Forward	CCGTGACATCAAGGAGAAGCT	58	201
	Reverse	TCGTGGATACCGCAAGATTCC	58	

The *Mt2* mRNA transcription in the hepatocytes was significantly down-regulated by different concentrations of MPs, whereas *Ces2* mRNA transcription was significantly increased. Although MPs at 175 μg L^{-1} had no significant effect on *Mt1* and *P450* mRNA transcription, *Mt1* and *P450* expression were meaningfully increased following exposure of fish to 350–1400 μg L^{-1} MPs (Fig. 3).

Exposure of fish to MPs at 350–1400 μg L^{-1} increased significantly SOD, GPx activities, MDA, and carbonyl protein levels in the hepatocytes. In contrast, G6PDH, GR, and CAT activities, glycogen, and total protein levels decreased significantly after 30 days' exposure to 350–1400 μg L^{-1} MPs. Furthermore, no significant changes were seen in the oxidative biomarkers after 30 days' exposure to 175 μg L^{-1} MPs (Table 3).

The plasma enzyme activities in fish after exposure to MPs are presented in Table 4. Results displayed that AST, ALT, GGT, ALP, LDH, BChE, and CPK activities in the plasma of *C. carpio* were significantly ($P < 0.05$) increased following exposure to 350–1400 μg L^{-1} MPs. However, no significant changes in these enzyme activities were detected in fish challenged with 175 μg L^{-1} MPs after 30 days.

Although no significant change was found in the albumin levels, we recorded a significant decrease in the total protein and globulins in MPs treated groups (350–1400 μg L^{-1}) at the end of the 30th day. Cholesterol and triglyceride contents increased in plasma of *C. carpio* exposed to 700 and 1400 μg L^{-1} MPs for 30 days. Moreover, the results presented in Table 4 show a significant increase in the creatinine levels in the plasma of fish treated with 350–1400 μg L^{-1} MPs.

4. Discussion

Although MPs were undetectable in some samples in fish exposed to 175 μg L^{-1}, the presence of MPs in the liver and intestine was confirmed at higher concentrations (350–1400 μg L^{-1}). The changes in FTIR transmittance (%) at the 1445–1650 wavenumber (cm^{-1}) indicated the presence of an aromatic ring (aryl) in the liver and intestinal tissue samples (Li et al., 2022a, 2022b). MPs and nano-plastics (NPs) may enter cells through pinocytosis or phagocytosis. Furthermore, the aryl hydrocarbon receptor (AhR) plays an important role in the uptake of plasticizers such as bisphenol A, 4-n-nonylphenol, phthalates, etc., and MPs into cells (Krüger et al., 2008). Gulmine et al. (2002) found that the bands set for polyethylene detection were in the regions 3000–2800, 1550–1400, and 750–650 cm^{-1}.

During the trial, the fish looked healthy and ate well. Furthermore, no mortality was observed in the tested fish. Results showed that the mRNA levels of *Mt1* significantly increased in the hepatocytes of *C. carpio* exposed to 350, 700, and 1400 μg L^{-1} of MPs (Fig. 3). However, the mRNA levels of Met1 significantly decreased after fish exposure to MPs. Derikvandy et al. (2020) showed that metallothionein-1 is often associated with the detoxification of metal ions, while metallothionein-2 is involved in copper ion homeostasis. Moreover, Santos et al. (2021) revealed that metallothioneins play an essential role in the neutralization of ROS in cells. Therefore, Increased *Mt1* gene expression may be a defense mechanism for scavenging ROS. Thus, increased *Mt1* gene expression may lead to increased metallothionein-1 levels, which may be a physiological response to the toxicity of MPs (Huang et al., 2021). Banaee et al., 2015 reported that exposure of aquatic animals to xenobiotic might increase metallothionein levels. However, decreased mRNA levels of *Mt2* may impair copper homeostasis in fish exposed to MPs. Decreased expression of the *Mt2* gene in zebrafish (*D. rerio*) exposed to MPs alone and in combination with copper was reported (Santos et al., 2021).

As soon as the MPs as external ligands bond with the AhR, the expression of the cytochrome P450 gene, the enzymes involved in detoxification, increases. As a result, the level of reactive oxygen species

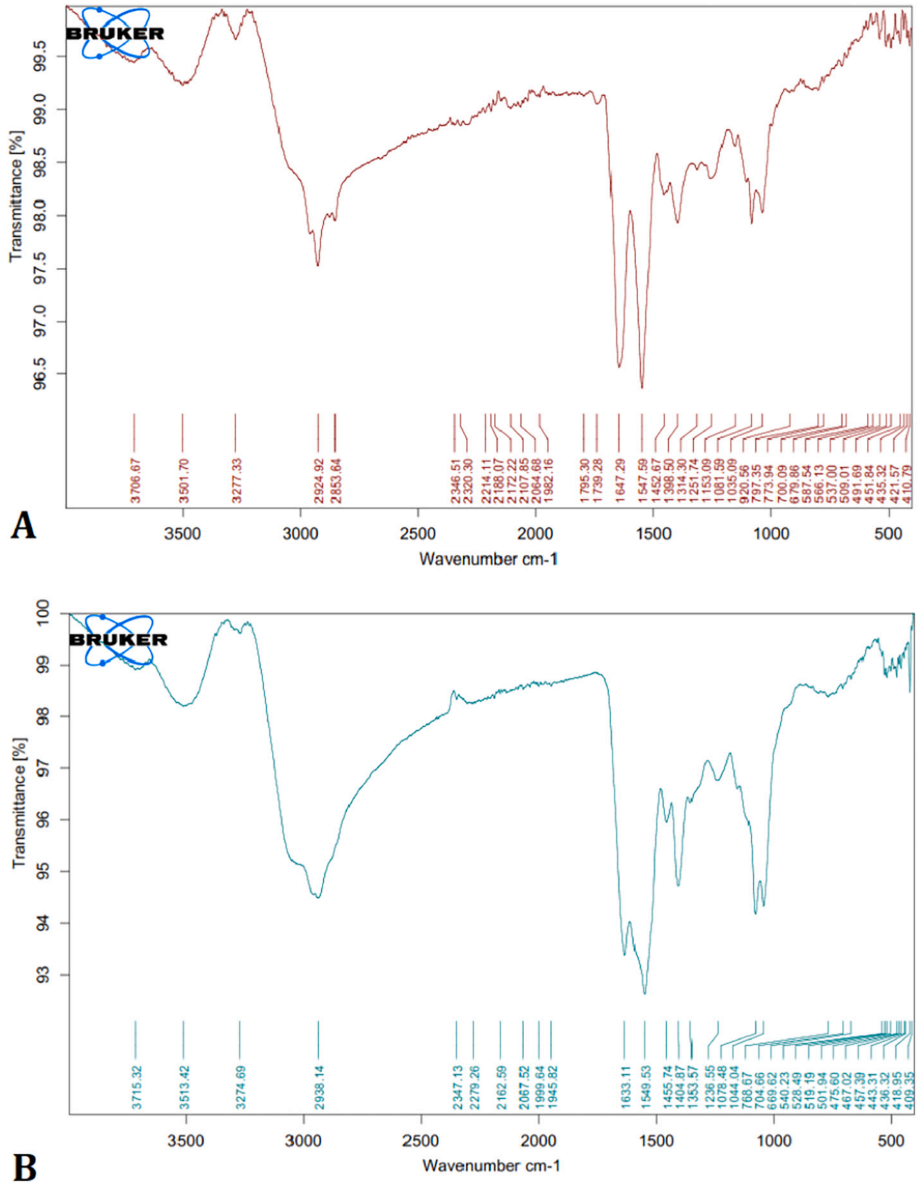

Fig. 1. FTIR diagram of polyethylene in the liver of *C. carpio* exposed to MPs. Fig. (A): liver (350 μg L^{-1} MPs); fig. (B): liver (1400 g μL^{-1} MPs).

(ROS) will increase as soon as phase one detoxification enzymes are activated (Krüger et al., 2008). Thus, the biochemical homeostasis of the blood can be disrupted as the rate of peroxidation of vital macromolecules increases. Fish exposure to 700 and 1400 μg L^{-1} of MPs increased *P450* gene expression levels. Gene expression level of *Ces2* increased significantly with increasing MPs concentration (Fig. 3). The induction of P450 and Ces3 genes in fish indicate that the fish detoxification system is involved in the metabolism of MPs. These data displayed that carboxylesterase and cytochrome P450 A might play a vital role in the detoxification of MPs. The activity of cytochrome P450 enzymes in the metabolism of xenobiotics can play an influential role in ROS production. Limonta et al. (2021) showed that zebrafish (*D. rerio*) exposure to MPs increased CYP450 gene expression. They stated that the induction of phase I detoxification enzymes such as CYP450 and Ces2 in fish showed that MPs could also stimulate the detoxification system (Limonta et al., 2021). Similarly, an increase in mRNA levels of CYP1A1 was reported in zebrafish (*D. rerio*) exposed to MPs (Xu et al., 2021).

Once MPs penetrate the bloodstream, they may enter cells through endocytosis. Li et al. (2022a, 2022b) showed that MPs could be absorbed by cells through endocytosis, leading to reactive oxygen species (ROS) production by NADPH oxidase. As a result, an increase in ROS can stimulate the antioxidant system. After 30 days, a significant enhancement was recorded in the SOD activity in the liver cells of *C. carpio* exposed to 350, 700, and 1400 μg L^{-1} MPs (Table 3). Elevated levels of superoxide anions may have increased SOD activity. A considerable increment was detected in the SOD activity in the bay scallop, *Argopecten irradians* (Song et al., 2020), red tilapia (*Oreochromis niloticus*) (Ding et al., 2018), zebrafish (*Danio rerio*) (Lu et al., 2016), and discus fish (*Symphysodon aequifasciatus*) (Wen et al., 2018) exposed to MPs.

Results showed a significant increase in hepatocyte GPx activity after exposure to 350, 700, and 1400 μg L^{-1} MPs (Table 3). This increase in GPx activity may accelerate the conversion of H_2O_2 and other peroxide radicals to H_2O and O_2 and may reduce ROS accumulation in tissues (Li et al., 2022a, 2022b). Moreover, this result may indicate that a significant increase in GPx activity is sufficient to neutralize ROS production in hepatocytes of fish exposed to MPs (Ribeiro et al., 2017).

In this study, CAT activity was declined in the liver of *C. carpio* treated with MPs (Table 3). This is probably due to overproduction of ROS or altered gene expression (Song et al., 2020), which is consistent with the findings of another study (Umamaheswari et al., 2021).

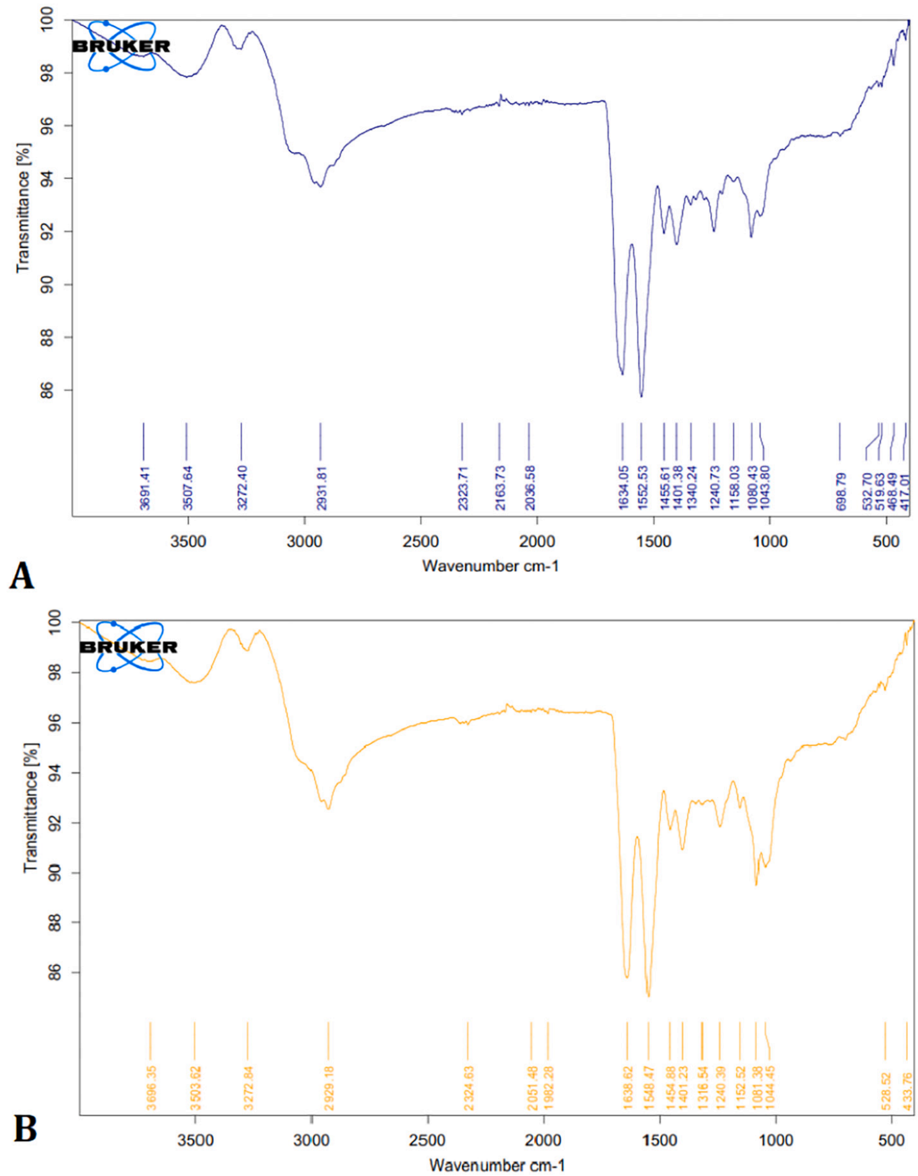

Fig. 2. FTIR diagram of polyethylene in the intestine of *C. carpio* exposed to MPs. Fig. (A): intestine (350 μg L^{-1} MPs); fig. (B): intestine (1400 g μL^{-1} MPs).

Table 2
Main transmittance of polyethylene in the IR region and their assignment (Li et al., 2022a, 2022b).

Group frequency, wavenumber	Assigned wavenumber	Assignment	Possible products
3620–3540	3566	O—H	Tertiary alcohol, OH stretch
2935–2915	2929	C—H	Methylene C—H asymmetric stretch
3100–2600	2821	C—H	Hydrocarbon
1680–1620	1639	C=C	Aromatic ring stretch
1650–1450	1508	Aromatic ring	Aromatic ring
1485–1445	1461	=CH_2	Aromatic ring (aryl)
1300–950	1172	C=O	Phenols, alcohols, ethers
1225–950	1092	C—H	Aromatics

Umamaheswari et al. (2021) found that exposure of fish to MPs decreased CAT gene expression. Moreover, different responses to SOD and CAT may indicate changes in intracellular ROS contents exposed to MPs. Increasing GPx activity may reduce H_2O_2 levels, and this phenomenon may inhibit the CAT activity. A significant decrease was observed in the CAT activity in wedge clam *Donax trunculus* (Tlili et al., 2020), and *Litopenaeus vannamei* (Hsieh et al., 2021) exposed to MPs. Also, Sayed et al. (2021) found that CAT activity was inhibited after African catfish (*Clarias gariepinus*) exposure to MPs.

However, the activities of GR and G6PDH showed a significant decrease in the hepatocytes of fish exposed to 350, 700, and 1400 μg L^{-1} MPs (Table 3). Glucose-6-phosphate dehydrogenase (G6PDH) is a vital enzyme in the biosynthesis of nicotinamide adenine dinucleotide phosphate (NADPH) in the pentose phosphate (PPP) pathway. Therefore, reduction of G6PDH activity can lead to a decrease in the level of NADPH biosynthesis. As a result, decreased NADPH levels could reduce GR activity. The results of this study also fully agree with this proof. The results also showed that a decrease in G6PDH activity could be associated with decrease in GR activity. GR can convert the oxidized form of glutathione (GSSG) to the reduced form (GSH) in the presence of NADPH and Flavin adenine dinucleotide (FAD). Therefore, decreased GR activity may reduce the cellular GSH levels and the free radical scavenging capability. As a result, the risk of oxidative stress in fish exposed to MPs

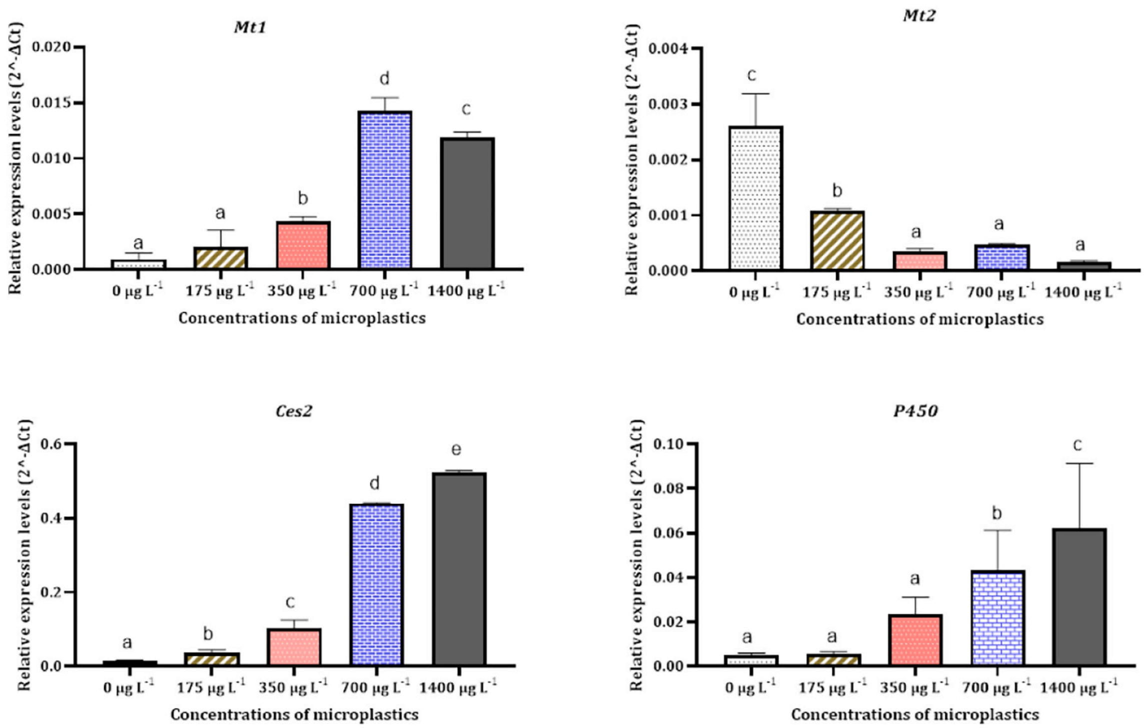

Fig. 3. Changes in the expression level of the *Mt1*, *Mt2*, *P450*, and *Ces2* genes in the hepatocytes of common carp (*C. carpio*) after exposure to different concentrations of MPs. Results are showed as mean ± standard deviation (S.D). Significant differences between different experimental groups were shown alphabetically ($p < 0.05$).

Table 3
Changes of antioxidant biomarkers in the hepatocytes of common carp (*C. carpio*) after exposure to different concentrations of MPs.

Oxidative biomarkers	Concentrations of MPs in the water				
	0.0 μg L^{-1}	175 μg L^{-1}	350 μg L^{-1}	700 μg L^{-1}	1400 μg L^{-1}
SOD (U g^{-1} protein)	1.8 ± 0.2^{a}	1.9 ± 0.2^{a}	2.2 ± 0.1^{b}	2.3 ± 0.2^{b}	2.4 ± 0.1^{b}
G6PDH (U g^{-1} protein)	16.9 ± 1.9^{b}	15.2 ± 1.7^{b}	10.8 ± 0.8^{a}	10.4 ± 0.5^{a}	9.5 ± 0.2^{a}
GPx (U g^{-1} protein)	7.3 ± 0.6^{a}	7.7 ± 0.5^{a}	8.9 ± 0.5^{b}	9.2 ± 0.5^{b}	12.2 ± 1.5^{c}
GR (U g^{-1} protein)	11.6 ± 0.9^{b}	10.9 ± 0.8^{b}	8.8 ± 0.5^{a}	8.5 ± 0.3^{a}	8.2 ± 0.1^{a}
CAT (KU g^{-1} protein)	20.1 ± 2.1^{b}	18.2 ± 1.3^{b}	14.5 ± 0.6^{a}	14.6 ± 1.0^{a}	13.9 ± 1.1^{a}
MDA (μmol g^{-1} tissue)	0.20 ± 0.01^{a}	0.32 ± 0.02^{b}	0.59 ± 0.02^{c}	0.60 ± 0.04^{c}	0.64 ± 0.03^{c}
Protein carbonyl (μmol g^{-1} protein)	0.53 ± 0.02^{a}	0.64 ± 0.01^{b}	0.92 ± 0.05^{c}	0.95 ± 0.05^{c}	1.14 ± 0.02^{d}
Total antioxidant (μmol g^{-1} protein)	8.76 ± 0.66^{c}	7.79 ± 0.45^{b}	5.65 ± 0.26^{a}	5.56 ± 0.21^{a}	5.08 ± 0.12^{a}
Glycogen (μmol g^{-1} tissue)	21.94 ± 1.48^{c}	21.27 ± 1.21^{c}	19.04 ± 0.63^{b}	18.60 ± 0.59^{b}	16.73 ± 0.63^{a}

Results are showed as mean ± standard deviation (S.D). Significant differences between different experimental groups were shown alphabetically ($p < 0.05$).

may increase following a decrease in GR and G6PDH activities. Furthermore, GSH is a cofactor to maintain GPx activity. Therefore, reducing cellular GSH levels can also negatively affect GPx function in the long run.

Imbalance between the antioxidant defense system and intracellular ROS can lead to oxidative stress (Ibrahim et al., 2019; Hamed et al., 2021b). Decreased total antioxidant capacity increases the chance of oxidative stress production (Ibrahim et al., 2021). Results revealed that total antioxidant levels were reduced in fish exposed to MPs. Decreased

Table 4
Changes in the blood biochemical parameters of common carp (*C. carpio*) after exposure to different concentrations of MPs.

Blood biochemical parameters	Concentrations of MPs in the water				
	0.0 μg L^{-1}	175 μg L^{-1}	350 μg L^{-1}	700 μg L^{-1}	1400 μg L^{-1}
AST (U L^{-1})	70.1 ± 4.7^{a}	73.3 ± 4.5^{a}	85.6 ± 3.5^{b}	88.6 ± 3.9^{b}	117.2 ± 3.8^{c}
ALT (U L^{-1})	22.3 ± 2.5^{a}	24.2 ± 2.9^{a}	30.8 ± 3.8^{b}	32.2 ± 4.1^{b}	34.7 ± 3.1^{b}
GGT (U L^{-1})	12.1 ± 0.9^{a}	13.3 ± 1.1^{a}	16.8 ± 1.3^{b}	17.4 ± 1.2^{b}	18.6 ± 1.2^{b}
ALP (U L^{-1})	73.6 ± 4.2^{a}	74.3 ± 3.1^{a}	86.4 ± 2.9^{b}	91.8 ± 2.5^{c}	101.6 ± 2.8^{d}
LDH (U L^{-1})	297.8 ± 46.8^{a}	311.3 ± 42.6^{ab}	350 ± 24.3^{abc}	355.5 ± 18.9^{bc}	379.0 ± 10.5^{c}
BChE (U L^{-1})	626.8 ± 10.6^{a}	650.0 ± 5.6^{ab}	679.6 ± 12.0^{b}	744.8 ± 22.6^{c}	867.5 ± 28.4^{d}
CPK (U L^{-1})	456.6 ± 30.0^{a}	460.6 ± 31.3^{ab}	470.5 ± 29.1^{ab}	471.4 ± 26.8^{ab}	507.4 ± 25.0^{b}
Total protein (g dL^{-1})	4.21 ± 0.10^{d}	4.16 ± 0.07^{cd}	4.00 ± 0.08^{bc}	3.98 ± 0.11^{ab}	3.82 ± 0.11^{a}
Albumin (g dL^{-1})	2.33 ± 0.06^{a}	2.32 ± 0.06^{a}	2.32 ± 0.05^{a}	2.32 ± 0.09^{a}	2.25 ± 0.12^{a}
Globulin (g dL^{-1})	1.89 ± 0.07^{c}	1.83 ± 0.07^{c}	1.68 ± 0.05^{b}	1.66 ± 0.03^{ab}	1.56 ± 0.05^{a}
Glucose (mg dL^{-1})	70.52 ± 7.84^{a}	75.96 ± 2.85^{a}	85.26 ± 3.48^{b}	92.66 ± 3.56^{bc}	99.35 ± 4.92^{c}
Triglycerides (mg dL^{-1})	158.9 ± 15.2^{a}	163.3 ± 13.9^{ab}	179 ± 12.4^{ab}	182.5 ± 14.8^{b}	183.7 ± 11.6^{b}
Cholesterol (mg dL^{-1})	169.1 ± 18.1^{a}	171.2 ± 17.5^{ab}	186.7 ± 10.5^{abc}	192.3 ± 11.6^{bc}	197.9 ± 4.9^{c}
Creatinine (mg dL^{-1})	0.75 ± 0.03^{a}	0.77 ± 0.03^{a}	0.84 ± 0.03^{b}	0.85 ± 0.02^{b}	0.99 ± 0.04^{c}

Results are showed as mean ± standard deviation (S.D). Significant differences between different experimental groups were shown alphabetically ($p < 0.05$).

total antioxidants combined with increased carbonyl group and MDA content suggest that exposure of fish to MPs can contribute to oxidative stress (Table 3). A significant reduction in total antioxidants was reported in African catfish (*C. gariepinus*) exposed to MPs (Sayed et al., 2021).

Compared with the reference group, MDA and carbonyl protein contents increased meaningfully in the hepatocytes after fish exposure to 350, 700, and 1400 μg L^{-1} MPs (Table 3). Peroxidation of vital macromolecules such as lipids and proteins can generate metabolites such as MDA and carbonyl protein derivatives (Solomando et al., 2020). Therefore, increased MDA and carbonyl groups in fish exposed to MPs can be a biomarker in diagnosing oxidative stress and failure of the antioxidant defense system. Increased MDA and carbonyl groups were observed in gilthead seabream (*Sparus aurata*) exposed to MPs (Solomando et al., 2020).

Glycogen levels in the hepatocytes in fish treated with 700 and 1400 μg L^{-1} MPs was considerably lower than the control group (Table 3). Decreased glycogen storage in hepatocytes can be attributed to its breakdown as an energy supply to counteract the cytotoxic effects of MPs. In fish treated with MPs, glycogen may break down into glucose, which is consistent with an increase in blood glucose. Japanese Medaka (*Oryzias latipes*) exposure to MPs decreased hepatic glycogen levels (Mallik et al., 2021).

A significant increase was recorded in AST and ALT activities in the plasma of fish challenged with 350, 700, and 1400 μg L^{-1} MPs. AST and ALT are cytoplasmic and mitochondrial enzymes (Table 4). Lipid peroxidation and destruction of cell membranes can cause AST and ALT to leak into the bloodstream. Therefore, increased AST and ALT activities may be due to membrane damage after the cytotoxicity of MPs. The activity of AST and ALT was increased in the serum of tilapia (*Oreochromis niloticus*) (Hamed et al., 2019), and *C. carpio* (Nematdoost Haghi and Banaee, 2017; Banaee et al., 2019c) after exposure to MPs caused by damage to the liver cell membrane. Plasticizers can play an influential role in cytotoxicity and increase AST and ALT activities (Banaee et al., 2021). The activity of AST significantly increased in the serum of sea bream (*Sparus aurata* L.) treated with MPs (Espinosa et al., 2017).

A meaningful increase in GGT activity was detected in the plasma of *C. carpio* treated with MPs. GGT is a membrane-bound enzyme that participate in the renovation and biodegradation of glutathione and detoxification from xenobiotic (Hatami et al., 2019). Therefore, the activity of GGT is essential in providing amino acids (glutamate, cysteine and glycine) for the synthesis and renewal of intracellular glutathione (Banaee et al., 2019a; Hatami et al., 2019). Furthermore, increased GGT activity indicates cellular glutathione depletion, especially in hepatocytes, leading to oxidative stress (Banaee et al., 2019b). Therefore, an enhanced GGT activity in the plasma of *C. carpio* after exposure to MPs can show oxidative stress. Damage to the liver cell membranes of fish exposed to MPs could lead to changes in plasma GGT activity (Banaee et al., 2019c).

After challenge with MPs, ALP activity in the plasma of *C. carpio* increased significantly. ALP is a membrane-bound metalloenzyme involved in protein phosphorylation, cell growth, programmed cell death, and cell migration (Banaee, 2020). Therefore, alterations in ALP activity may be related to cell membrane damage (Banaee, 2020; Banaee et al., 2019b). Exposure of *C. carpio* to MPs increased ALP activity (Nematdoost Haghi and Banaee, 2017; Banaee et al., 2019c). Lipid peroxidation and changes in the structure of cellular phospholipid layers can lead to the release of ALP into the blood of fish treated with MPs (Banaee et al., 2019c). Increased ALP activity in the blood can indicate programmed cell death (Banaee, 2020). Increased ALP activity was reported in the serum of *O. niloticus* (Hamed et al., 2019), and *Symphysodon aequifasciatus* (Wen et al., 2018) after challenge with MPs.

Results showed that LDH activity in fish treated with 700 and 1400 μg L^{-1} MPS was significantly higher than the reference group. LDH plays an essential role in converting lactate to pyruvate, NAD+ to NADH and vice versa (Banaee et al., 2020; Soleimany et al., 2016). Hypoxia conditions and mitochondrial oxidation dysfunction could increase LDH activity in the blood. Moreover, increased LDH activity can be due to necrosis or cell death (Banaee et al., 2020). The release of LDH into the blood can be related to the destruction of the cell membrane structure. Increased LDH activity in the blood of carp (*C. carpio*) exposed to MPs and cadmium has been reported (Banaee et al., 2019c). Similar results have been observed in *C. carpio* treated with MPs and paraquat (Nematdoost Haghi and Banaee, 2017). Wang et al. (2020) recorded an enhancement in LDH activity in discus fish following challenge with MPs (Wen et al., 2018). This increase may be the result of cellular toxicity due to plastic polymer emollients (Beltifa et al., 2018).

The highest activity of CPK was detected in the fish exposed to 1400 μg L^{-1} MPs. CPK is found in large amounts in muscle cells, heart tissue, gills, kidneys, and brain of animals, which can be released into the bloodstream due to cell damage (Perrault et al., 2016). Increased creatinine kinase activity can be attributed to injury to the muscles and kidneys of fish. CPK activity was increased in sea bream (*Sparus aurata* L.) after 15 days of feeding with diets contaminated with polyvinyl chloride MPs (Espinosa et al., 2017). Increased creatinine kinase in the blood of freshwater turtle (*Emys orbicularis*) exposed to MPs (polyethylene) has also been reported (Banaee et al., 2021). Increased CPK in turtle blood may be due to oxidative stress hurt to muscle cells and kidneys (Espinosa et al., 2017).

Fish exposure to 350, 700 and 1400 g μL^{-1} MPs increased BChE activity. BChE is found in all tissues and cells, including erythrocytes (Han et al., 2019). BChE in red blood cells contributes to the transmission of cell signals. Oxidative stress and lipid peroxidation could mitigate BChE activity (Banaee et al., 2019b) in fish challenged with MPs (Barboza and Guilhermino, 2018). Changes in BChE activity can affect the half-life of erythrocytes (Banaee et al., 2019b; Hatami et al., 2019). Elevated plasma BChE may indicate hemolysis of red blood cells and release of this enzyme into the bloodstream.

The results for biochemical parameters are presented in Table 4. There were no notable changes in the albumin contents between all experimental groups. A significant decrease was observed in the total protein and globulin contents in the plasma of fish following challenge with 700 and 1400 μg L^{-1} MPs. The ingestion of MPs can cause malnutrition and disruption of energy budgeting (Gray et al., 2022). Jin et al. (2019) found that MPs could affect the uptake of nutrients, including amino acids (Jin et al., 2019). Decreased amino acid absorption may be due to physical or oxidative damage to the intestinal epithelium. Furthermore, the interaction between plasticizers with serum proteins could alter the structure and function of proteins (De-la-Torre, 2020). Significant reductions in total protein, albumin, and globulin were observed in the blood of fish exposed to MPs (Nematdoost Haghi and Banaee, 2017; Banaee et al., 2019c).

Glucose levels increased significantly in fish treated with 350, 700, and 1400 μg L^{-1} MPS. Blood glucose concentration is maintained through homeostasis, creating a balance between glucose production rate and glucose storage as glycogen (Kim et al., 2021). Elevated blood glucose levels may be a physiological mechanism to energy supply to neutralize the toxic effects of plasticizers (Banaee et al., 2021). Increased blood glucose was reported in freshwater turtles (Banaee et al., 2021), tilapia *O. niloticus* (Hamed et al., 2019), and *C. carpio* after exposure to MPs. Corticosteroids can control changes in glucose levels under various physiological conditions (Zanuzzo et al., 2019). Therefore, an increase in blood glucose may be due to MPs following increased corticosteroids in fish. Impaired blood glucose homeostasis may be related to damage to the pancreatic beta cells (Ropero et al., 2008). The changes in the glucose contents may be related to the toxicity effect of plasticizers on endocrine secretions and enzyme activities that control blood glucose (Xu et al., 2018). Zhou et al. (2015) and Stojanoska et al. (2017) found that plasticizers could interrupt the cellular signaling pathway regulating blood glucose metabolism (Zhou et al., 2015; Stojanoska et al., 2017).

A major increment in creatinine levels was detected in MP-treated fish compared to the reference group. The breakdown of creatine phosphate during muscle and protein metabolism results in producing an excretory metabolite called creatinine, which the kidneys of fish must excrete. Therefore, if the kidneys do not do their job properly, the level of creatinine in the blood increases. In fact, blood creatinine level, like urea, is an excellent biomarker to assess the glomerular filtration rate. Therefore, increased creatinine levels in the blood of freshwater turtles, *E. orbicularis* (Banaee et al., 2021), and tilapia, *O. niloticus* (Hamed et al., 2019) exposed to MPs may be related to changes in glomerular filtration.

Cholesterol and triglyceride levels in fish exposed to 700 and 1400 μg L^{-1} MPs were significantly higher than the reference group. Alterations in the function of enzymes and hormones involved in lipid metabolism may increase cholesterol and triglycerides in the blood of fish challenged with MPs. Cholesterol is an essential component of cellular phospholipid membranes that can also be used as an energy source (Brandts et al., 2021). Furthermore, cholesterol plays a critical role in cell signaling (Brandts et al., 2021). The liver regulates the metabolism of lipids, including cholesterol, and exposure to MPs can affect lipids in the circulatory system (Banaee et al., 2019c). Changes in fish plasma cholesterol can affect cell membrane structures and energy budgeting (Kim et al., 2021). Brandts et al. (2021) observed a significant increase in blood cholesterol in *Sparus aurata* exposed to MPs.

Consumption of MPs can affect the ratio of triglycerides and cholesterol in fish blood (Jovanović, 2017). The change in the lipid profiles could be related to dysfunction in the turnover of lipids and deficiency biosynthesis of lipoproteins (Banaee et al., 2019c). Overexpression of essential genes in lipid biosynthesis in the liver of mouse treated with plasticizers can effectively change the concentration of triglycerides and cholesterol in the blood (Ke et al., 2016).

5. Conclusion

FTIR data indicated that MPs were found in the liver and intestine *C. carpio* exposed to MPs. Furthermore, changes in the expression of enzymes and proteins involved in detoxification indicate that MPs are toxic to fish. The response of the detoxification system of fish hepatocytes exposed to MPs was observed as a change in *MT*-1, *MT*-2, *CES2*, and *P450* mRNA expression. Exposure of *C. carpio* to MPs for 30 days led to a collapse of the antioxidant defense system, was indicated by a change in CAT, SOD, GPx, G6PDH, and GR activities and total antioxidant capacity, MDA, and carbonyl protein contents. Results showed that alterations in oxidative stress biomarkers have shown that MPs could disturb the balance between antioxidants and intracellular oxidants by producing ROS. Changes in oxidative stress biomarkers can confirm the results observed in biochemical parameters (AST, ALT, ALP, LDH, GGT, and CPK activities), all of which occur following damage to cell membranes. Disruption of biochemical homeostasis also indicated cell damage and disruption of cellular metabolic processes. Results could confirm the hypothesis of this study. In conclusion, this study revealed that MPs could be a severe threat to aquatic life, even in minimal concentrations.

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CRediT authorship contribution statement

Mehdi Banaei; Investigation, Project administration, Validation, Formal analysis, Writing - Original Draft

Mohsen Forouzanfar; Supervisor, Validation

Mojtaba Jafarinia: Adviser, Validation

Declaration of competing interest

The authors declare that they have no conflict of interest.

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Ethical considerations

Compliance with ethical guidelines

In this study, none of the authors used human beings as research subjects. International, national and institutional guidelines for the care and use of animals were followed. Experimental protocols were done following the Iranian animal ethics framework under the supervision of the Iranian Society for the Prevention of Cruelty to Animals and Islamic Azad University, Marvdasht Branch (162251822-IAU).

References

Banaee, M., 2020. Alkaline phosphatase activity as a biochemical biomarker in aqua-toxicological studies. Int. J. Aquat. Biol. 8 (2), 143–147. https://doi.org/10.22034/ijab.v8i2.880.

Banaee, M., Akhlaghi, M., Soltanian, S., Gholamhosseini, A., Heidarieh, H., Fereidouni, M.S., 2019. Acute exposure to chlorpyrifos and glyphosate induces changes in hemolymph biochemical parameters in the crayfish, Atacus leptodactylus (Eschscholtz, 1823). Comp. Biochem. Physiol. C: Toxicol. Pharmacol. 222, 145–155. https://doi.org/10.1016/j.cbpc.2019.05.003.

Banaee, M., Akhlaghi, M., Soltanian, S., Sureda, A., Gholamhosseini, A., Rakhshaninejad, M., 2020. Combined effects of exposure to sub-lethal concentration of the insecticide chlorpyrifos and the herbicide glyphosate on the biochemical changes in the freshwater crayfish pontastacus leptodactylus. Ecotoxicology 29 (9), 1500–1515. https://doi.org/10.1007/s10646-020-02233-0.

Banaee, M., Gholamhosseini, A., Sureda, A., Soltanian, S., Fereidouni, M.S., Ibrahim, A. T., 2021. Effects of microplastic exposure on the blood biochemical parameters in the pond turtle (Emys orbicularis). Environ. Sci. Pollut. Res. 28 (8), 9221–9234. https://doi.org/10.1007/s11356-020-11419-2.

Banaee, M., Mohammadipour, S., Madhani, S., 2015. Effects of sublethal concentrations of permethrin on bioaccumulation of cadmium in zebra cichlid (Cichlasoma nigrofasciatum). Toxicol. Environ. Chem. 97 (2), 200–207. https://doi.org/10.1080/02772248.2015.1031668.

Banaee, M., Soltanian, S., Sureda, A., Gholamhosseini, A., Haghi, B.N., Akhlaghi, M., Derikvandy, A., 2019c. Evaluation of single and combined effects of cadmium and micro-plastic particles on biochemical and immunological parameters of common carp (Cyprinus carpio). Chemosphere 236, 124335. https://doi.org/10.1016/j.chemosphere.2019.07.066.

Banaee, M., Tahery, S., Nematdoost Haghi, B., Shahafve, S., Vaziriyan, M., 2019b. Blood biochemical changes in common carp (Cyprinus carpio) upon co-exposure to titanium dioxide nanoparticles and paraquat. Iran. J. Fish. Sci. 18 (2), 242–255. https://doi.org/10.22092/ijfs.2019.118174.

Barboza, L.G., Guilhermino, L., 2018. Microplastics cause neurotoxicity, oxidative damage and energy-related changes and interact with the bioaccumulation of mercury in the European seabass, Dicentrarchus labrax (linnaeus, 1758). Aquat. Toxicol. 195, 49–57. https://doi.org/10.1016/j.aquatox.2017.12.008.

Beltifa, A., Feriani, A., Macherki, M., Ghorbel, A., Ghazouani, L., Di Bella, G., Mansour, H.B., 2018. Persistent plasticizers and bisphenol in the cheese of Tunisian markets induced biochemical and histopathological alterations in male BALB/c mice. Environ. Sci. Pollut. Res. 25 (7), 6545–6557. https://doi.org/10.1007/s11356-017-0857-6.

Benzie, I., Strain, J., 1996. The ferric reducing ability of plasma (FRAP) as a measure of "antioxidant power: the FRAP assay. Anal. Biochem. 239, 70–76. https://doi.org/10.1006/abio.1996.0292.

Brandts, I., Barría, C., Martins, M.A., Franco-Martínez, L., Barreto, A., Tvarijonaviciute, A., Teles, M., 2021. Waterborne exposure of gilthead seabream (Sparus aurata) to polymethylmethacrylate nanoplastics causes effects at cellular and molecular levels. J. Hazard. Mater. 403, 123590 https://doi.org/10.1016/j.jhazmat.2020.123590.

Colombo, G., Clerici, M., Garavaglia, M.E., Giustarini, D., Rossi, R., Milzani, A., Dalle-Donne, I., 2016. A step-by-step protocol for assaying protein carbonylation in biological samples. J. Chromatogr. B 1019, 178–190. https://doi.org/10.1016/j.jchromb.2015.11.052.

Critchell, K., Hoogenboom, M.O., 2018. Effects of microplastic exposure on the body condition and behaviour of planktivorous reef fish (Acanthochromis polyacanthus). PLoS One 13 (3), e0193308. https://doi.org/10.1371/journal.pone.0193308.

Cunha, C., Lopes, J., Paulo, J., Faria, M., Kaufmann, M., Nogueira, N., Cordeiro, N., 2020. The effect of microplastics pollution in microalgal biomass production: a biochemical study. Water Res. 186, 116370 https://doi.org/10.1016/j.watres.2020.116370.

De-la-Torre, G.E., 2020. Microplastics: an emerging threat to food security and human health. J. Food Sci. Technol. 57 (5), 1601–1608. https://doi.org/10.1007/s13197-019-04138-1.

Derikvandy, A., Pourkhabbaz, H.R., Banaee, M., Sureda, A., Haghi, N., Pourkhabbaz, A. R., 2020. Genotoxicity and oxidative damage in zebrafish (Danio rerio) after exposure to effluent from ethyl alcohol industry. Chemosphere 251, 126609. https://doi.org/10.1016/j.chemosphere.2020.126609.

Ding, J., Huang, Y., Liu, S., Zhang, S., Zou, H., Wang, Z., Geng, J., 2020. Toxicological effects of nano- and micro-polystyrene plastics on red tilapia: are larger plastic particles more harmless? J. Hazard. Mater. 396, 122693 https://doi.org/10.1016/j.jhazmat.2020.122693.

Ding, J., Zhang, S., Razanajatovo, R.M., Zou, H., Zhu, W., 2018. Accumulation, tissue distribution, and biochemical effects of polystyrene microplastics in the freshwater fish red tilapia (Oreochromis niloticus). Environ. Pollut. 238, 1–9. https://doi.org/10.1016/j.envpol.2018.03.001.

Dong, X., Zhu, L., Jiang, P., Wang, X., Liu, K., Li, C., Li, D., 2021. Seasonal biofilm formation on floating microplastics in coastal waters of intensified marinculture area. Mar. Pollut. Bull. 171, 112914 https://doi.org/10.1016/j.marpolbul.2021.112914.

Draper, H.H., Hadley, M., 1990. Malondialdehyde determination as index of lipid peroxidation. Methods Enzymol. 186, 421–431. https://doi.org/10.1016/0076-6879(90)86135-i.

Espinosa, C., Cuesta, A., Esteban, M.Á., 2017. Effects of dietary polyvinylchloride microparticles on general health, immune status and expression of several genes related to stress in gilthead seabream (Sparus aurata L.). Fish Shellfish Immunol. 68, 251–259. https://doi.org/10.1016/j.fsi.2017.07.006.

Frias, J.P., Gago, J., Otero, V., Sobral, P., 2016. Microplastics in coastal sediments from southern Portuguese shelf waters. Mar. Environ. Res. 114, 24–30. https://doi.org/10.1016/j.marenvres.2015.12.006.

Góth, L.A., 1991. Simple method for determination of serum catalase and revision of reference range. Clin. Chim. Acta 196, 143–152.

Gray, A.D., Weinstein, J.E., Riegerix, R.C., 2022. Assessment of acute toxicity and developmental transformation impacts of polyethylene microbead exposure on larval daggerblade grass shrimp (Palaemon pugio). Sci. Rep. 12 (1), 6967. https://doi.org/10.1038/s41598-022-10999-y.

Gulmine, J.V., Janissek, P.R., Heise, H.M., Akcelrud, L., 2002. Polyethylene characterization by FTIR. Polym. Test. 21 (5), 557–563. https://doi.org/10.1016/S0142-9418(01)00124-6.

Hamed, H.S., Ali, R.M., Shaheen, A.A., Hussein, N.M., 2021a. Chitosan nanoparticles alleviated endocrine disruption, oxidative damage, and genotoxicity of bisphenol-A-intoxicated female african catfish. Comp. Biochem. Physiol. C Toxicol. Pharmacol. 248, 109104 https://doi.org/10.1016/j.cbpc.2021.109104.

Hamed, H.S., Ismal, S.M., Faggio, C., 2021b. Effect of allicin on antioxidant defense system, and immune response after carbofuran exposure in Nile tilapia, Oreochromis niloticus. Comp. Biochem. Physiol. C Toxicol. Pharmacol. 240, 108919 https://doi.org/10.1016/j.cbpc.2020.108919.

Hamed, M., Soliman, H.A., Badrey, A.E., Osman, A.G., 2021c. Microplastics induced histopathological lesions in some tissues of tilapia (Oreochromis niloticus) early juveniles. Tissue Cell 71, 101512. https://doi.org/10.1016/j.tice.2021.101512.

Hamed, M., Soliman, H.A., Osman, A.G., Sayed, A.E., 2019. Assessment the effect of exposure to microplastics in Nile tilapia (Oreochromis niloticus) early juvenile: I. Blood biomarkers. Chemosphere 228, 345–350. https://doi.org/10.1016/j.chemosphere.2019.04.153.

Han, S.H., Park, J.C., Byun, M.S., Yi, D., Lee, J.H., Lee, D.Y., Mook-Jung, I., 2019. Blood acetylcholinesterase level is a potential biomarker for the early detection of cerebral amyloid deposition in cognitively normal individuals. Neurobiol. Aging 73, 21–29. https://doi.org/10.1016/j.neurobiolaging.2018.09.001.

Hatami, M., Banaee, M., Nematdoost Haghi, B., 2019. Sub-lethal toxicity of chlorpyrifos alone and in combination with polyethylene glycol to common carp (Cyprinus carpio). Chemosphere 219, 981–988. https://doi.org/10.1016/j.chemosphere.2018.12.077.

Hsieh, S.L., Wu, Y.C., Xu, R.Q., Chen, Y.T., Chen, C.W., Singhania, R.R., Dong, C.D., 2021. Effect of polyethylene microplastics on oxidative stress and histopathology damages in Litopenaeus vannamei. Environ. Pollut. 288, 117800 https://doi.org/10.1016/j.envpol.2021.117800.

Huang, W., Song, B., Liang, J., Niu, Q., Zeng, G., Shen, M., Zhang, Y., 2021. Microplastics and associated contaminants in the aquatic environment: a review on their ecotoxicological effects, trophic transfer, and potential impacts to human health. J. Hazard. Mater. 405, 124187 https://doi.org/10.1016/j.jhazmat.2020.124187.

Ibrahim, A.T., Banaee, M., Sureda, A., 2019. Selenium protection against mercury toxicity on the male reproductive system of Clarias gariepinus. Comp. Biochem. Physiol. C: Toxicol. Pharmacol. 225, 108583 https://doi.org/10.1016/j.cbpc.2019.108583.

Ibrahim, A.T., Banaee, M., Sureda, A., 2021. Genotoxicity, oxidative stress, and biochemical biomarkers of exposure to green synthesized cadmium nanoparticles in Oreochromis niloticus (L.). Comp. Biochem. Physiol. C: Toxicol. Pharmacol. 242, 108942 https://doi.org/10.1016/j.cbpc.2020.108.

Iheanacho, S.C., Igberi, C., Amadi-Eke, A., Chinonyerem, D., Iheanacho, A., Avwemoya, F., 2020. Biomarkers of neurotoxicity, oxidative stress, hepatotoxicity and lipid peroxidation in Clarias gariepinus exposed to melamine and polyvinyl chloride. Biomarkers 25 (7), 603–610. https://doi.org/10.1080/1354750X.2020.1821777.

Jin, Y., Lu, L., Tu, W., Luo, T., Fu, Z., 2019. Impacts of polystyrene microplastic on the gut barrier, microbiota and metabolism of mice. Sci. Total Environ. 649, 308–317. https://doi.org/10.1016/j.scitotenv.2018.08.353.

Johnson, A.M., Rohlfs, E.M., Silverman, L.M., 1999. Proteins. In: Burtis, C.A., Ashwood, E.R. (Eds.), Tietz Textbook of Clinical Chemistry, 3rd ed. W.B. Saunders Company, Philadelphia.

Johnson, G., Moore, S.W., 2012. Why has butyrylcholinesterase been retained? Structural and functional diversification in a duplicated gene. Neurochem. Int. 61 (5), 783–797. https://doi.org/10.1016/j.neuint.2012.06.016.

Jovanović, B., 2017. Ingestion of microplastics by fish and its potential consequences from a physical perspective. Integr. Environ. Assess. Manag. 13 (3), 510–515. https://doi.org/10.1002/ieam.1913.

Ke, Z.H., Pan, J.X., Jin, L.Y., Xu, H.Y., Yu, T.T., Ullah, K., Huang, H.F., 2016. Bisphenol a exposure may induce hepatic lipid accumulation via reprogramming the DNA methylation patterns of genes involved in lipid metabolism. Sci. Rep. 6, 31331. https://doi.org/10.1038/srep31331.

Kim, J.H., Yu, Y.B., Choi, J.H., 2021. Toxic effects on bioaccumulation, hematological parameters, oxidative stress, immune responses and neurotoxicity in fish exposed to microplastics: a review. J. Hazard. Mater. 413, 125423 https://doi.org/10.1016/j.jhazmat.2021.125423.

Krüger, T., Long, M., Bonefeld-Jørgensen, E.C., 2008. Plastic components affect the activation of the aryl hydrocarbon and the androgen receptor. Toxicology 246 (2–3), 112–123. https://doi.org/10.1016/j.tox.2007.12.028.

Li, J., Yao, X., Chen, S., Xu, K., Fan, B., Yang, D., Qiao, H., 2022. Investigation on the co-pyrolysis of agricultural waste and high-density polyethylene using TG-FTIR and artificial neural network modelling. Process Saf. Environ. Prot. 160, 341–353. https://doi.org/10.1016/j.psep.2022.02.033.

Li, Z., Chang, X., Hu, M., Fang, J.K., Sokolova, I.M., Huang, W., Wang, Y., 2022. Is microplastic an oxidative stressor? Evidence from a meta-analysis on bivalves. J. Hazard. Mater. 43 (B), 127211 https://doi.org/10.1016/j.jhazmat.2021.127211.

Limonta, G., Mancia, A., Abelli, L., Fossi, M.C., Caliani, I., Panti, C., 2021. Effects of microplastics on head kidney gene expression and enzymatic biomarkers in adult zebrafish. Comp. Biochem. Physiol. C: Toxicol. Pharmacol. 245, 109037 https://doi.org/10.1016/j.cbpc.2021.109037.

Livak, K.J., Schmittgen, T., 2001. Analysis of relative gene expression data using real-time quantitative PCR and the 2(-Delta Delta C(T)) method. Methods 25 (4), 402–408. https://doi.org/10.1006/meth.2001.1262.

Lu, Y., Zhang, Y., Deng, Y., Jiang, W., Zhao, Y., Geng, J., Ren, H., 2016. Uptake and accumulation of polystyrene microplastics in zebrafish (Danio rerio) and toxic effects in liver. Environ. Sci. Technol. 50 (7), 4054–4060. https://doi.org/10.1021/acs.est.6b00183.

Mallik, A., Xavier, K.A., Naidu, B.C., Nayak, B.B., 2021. Ecotoxicological and physiological risks of microplastics on fish and their possible mitigation measures. Sci. Total Environ. 779, 146433 https://doi.org/10.1016/j.scitotenv.2021.146433.

Moss, D.V., Henderson, A.R., 1999. Clinical enzymology. In: Burtis, C.A., Ashwood, E.R. (Eds.), Tietz Textbook of Clinical Chemistry, 3rd ed. W.B. Saunders Company, Philadelphia.

Mughini-Gras, L., van der Plaats, R.Q., van der Wielen, P.W., Bauerlein, P.S., de Roda Husman, A.M., 2021. Riverine microplastic and microbial community compositions: a field study in the Netherlands. Water Res. 192, 116852 https://doi.org/10.1016/j.watres.2021.116852.

Nematdoost Haghi, B., Banaee, M., 2017. Effects of micro-plastic particles on paraquat toxicity to common carp (Cyprinus carpio): biochemical changes. Int. J. Environ. Sci. Technol. 14 (3), 521–530. https://doi.org/10.1007/s13762-016.

Oliveira, M., Ribeiro, A., Hylland, K., Guilhermino, L., 2013. Single and combined effects of microplastics and pyrene on juveniles (0+ group) of the common goby Pomatoschistus microps (Teleostei, Gobiidae). Ecol. Indic. 34, 641–647. https://doi.org/10.1016/j.ecolind.2013.06.019.

Parker, B.W., Beckingham, B.A., Ingram, B.C., Ballenger, J.C., Weinstein, J.E., Sancho, G., 2020. , J. E., & , G. (2020). Microplastic and tire wear particle occurrence in fishes from an urban estuary: Influence of feeding characteristics on exposure risk. Mar. Pollut. Bull. 160, 111539 https://doi.org/10.1016/j.marpolbul.2020.111539.

Perrault, J.R., Bauman, K.D., Greenan, T.M., Blum, P.C., Henry, M.S., Walsh, C.J., 2016. Maternal transfer and sublethal immune system effects of brevetoxin exposure in nesting loggerhead sea turtles (Caretta caretta) from western Florida. Aquat. Toxicol. 180, 131–140. https://doi.org/10.1016/j.aquatox.2016.09.020.

Qu, H., Ma, R., Wang, B., Yang, J., Duan, L., Yu, G., 2019. Enantiospecific toxicity, distribution and bioaccumulation of chiral antidepressant venlafaxine and its metabolite in loach (Misgurnus anguillicaudatus) co-exposed to microplastic and the drugs. J. Hazard. Mater. https://doi.org/10.1016/j.jhazmat.2018.04.041. Article in Press.

Ribeiro, F., Garcia, A.R., Pereira, B.P., Fonseca, M., Mestre, N.C., Fonseca, T.G., Bebianno, M.J., 2017. Microplastics effects in Scrobicularia plana. Mar. Pollut. Bull. 122 (1–2), 379–391. https://doi.org/10.1016/j.marpolbul.2017.06.078.

Rifai, N., Bachorik, P.S., Albers, J.J., 1999. Lipids, lipoproteins and apolipoproteins. In: Burtis, C.A., Ashwood, E.R. (Eds.), Tietz Textbook of Clinical Chemistry, 3rd edition. W.B. Saunders Company, Philadelphia.

Ropero, A.B., Alonso-Magdalena, P., Quesada, I., Nadal, A., 2008. The role of estrogen receptors in the control of energy and glucose homeostasis. Steroids 73 (9–10), 874–879. https://doi.org/10.1016/j.steroids.2007.12.018.

Sacks, D.B., 1999. Carbohydrates. In: Burtis, C.A., Ashwood, E.R. (Eds.), Tietz Textbook of Clinical Chemistry, 3rd ed. W.B. Saunders Company, Philadelphia, pp. 766–785.

Santos, D., Félix, L., Luzio, A., Parra, S., Bellas, J., Monteiro, S.M., 2021. Single and combined acute and subchronic toxic effects of microplastics and copper in zebrafish (Danio rerio) early life stages. Chemosphere 277, 130262. https://doi.org/10.1016/j.chemosphere.2021.130262.

Sayed, A.E., Hamed, M., Badrey, A.E., Soliman, H.A., 2021. Bioremediation of hemotoxic and oxidative stress induced by polyethylene microplastic in Clarias gariepinus using lycopene, citric acid, and chlorella. Comp. Biochem. Physiol. C: Toxicol. Pharmacol. 250, 109189 https://doi.org/10.1016/j.cbpc.2021.109189.

Soleimany, V., Banaee, M., Mohiseni, M., Nematdoost Hagi, B., Mousavi Dehmourdi, L., 2016. Evaluation of pre-clinical safety and toxicology of Althaea officinalis extracts as naturopathic medicine for common carp (Cyprinus carpio). Iran. J. Fish. Sci. 15 (2), 613–629.

Solomando, A., Capó, X., Alomar, C., Álvarez, E., Compa, M., Valencia, J.M., Sureda, A., 2020. Long-term exposure to microplastics induces oxidative stress and a pro-inflammatory response in the gut of Sparus aurata linnaeus, 1758. Environ. Pollut. 266 (1), 115295 https://doi.org/10.1016/j.envpol.2020.115295.

Song, J.A., Choi, C.Y., Park, H.S., 2020. Exposure of bay scallop Argopecten irradians to micro-polystyrene: bioaccumulation and toxicity. Comp. Biochem. Physiol. C Toxicol. Pharmacol. 236, 108801 https://doi.org/10.1016/j.cbpc.2020.108801.

Stojanoska, M.M., Milosevic, N., Milic, N., Abenavoli, L., 2017. The influence of phthalates and bisphenol a on the obesity development and glucose metabolism disorders. Endocrine 55 (3), 666–681. https://doi.org/10.1007/s12020-016-1158-4.

Strungaru, S.A., Jijie, R., Nicoara, M., Plavan, G., Faggio, C., 2019. Micro- (nano) plastics in freshwater ecosystems: abundance, toxicological impact and quantification methodology. TrAC Trends Anal. Chem. 110, 116–128. https://doi.org/10.1016/j.trac.2018.10.025.

Sun, T., Zhan, J., Li, F., Ji, C., Wu, H., 2021. Evidence-based meta-analysis of the genotoxicity induced by microplastics in aquatic organisms at environmentally relevant concentrations. Sci. Total Environ. 783, 147076 https://doi.org/10.1016/j.scitotenv.2021.147076.

Tlili, S., Jemai, D., Brinis, S., Regaya, I., 2020. Microplastics mixture exposure at environmentally relevant conditions induce oxidative stress and neurotoxicity in the wedge clam donax trunculus. Chemosphere 258, 127344. https://doi.org/10.1016/j.chemosphere.2020.127344.

Umamaheswari, S., Priyadarshinee, S., Bhattacharjee, M., Kadirvelu, K., Ramesh, M., 2021. Exposure to polystyrene microplastics induced gene modulated biological responses in zebrafish (Danio rerio). Chemosphere 281, 128592. https://doi.org/10.1016/j.chemosphere.2020.128592.

Wang, W., Ge, J., Yu, X., 2020. Bioavailability and toxicity of microplastics to fish species: a review. Ecotoxicol. Environ. Saf. 189, 109913 https://doi.org/10.1016/j.ecoenv.2019.109913.

Watts, A.J., Lewis, C., Goodhead, R.M., Beckett, S.J., Moger, J., Tyler, C.R., 2014. Uptake and retention of microplastics by the shore crab Carcinus maenas. Environ. Sci. Technol. 48 (15), 8823–8830. https://doi.org/10.1021/es501090e.

Weber, A., von Randow, M., Voigt, A.L., von der Au, M., Fischer, E., Meermann, B., Wagner, M., 2021. Ingestion and toxicity of microplastics in the freshwater gastropod Lymnaea stagnalis: no microplastic-induced effects alone or in combination with copper. Chemosphere 263, 128040. https://doi.org/10.1016/j.chemosphere.2020.128040.

Wen, B., Jin, S., Chen, Z., Gao, J., Liu, Y., Liu, J., Feng, X., 2018. Single and combined effects of microplastics and cadmium on the cadmium accumulation, antioxidant defence and innate immunity of the discus fish (Symphysodon aequifasciatus). Environ. Pollut. 243, 462–471. https://doi.org/10.1016/j.envpol.2018.09.029.

Xu, J., Zhou, L., Wang, S., Zhu, J., Liu, T., Jia, Y., Ye, L., 2018. Di-(2-ethylhexyl)-phthalate induces glucose metabolic disorder in adolescent rats. Environ. Sci. Pollut. Res. Int. 25 (4), 3596–3607. https://doi.org/10.1007/s11356-017-0738-z.

Xu, K., Zhang, Y., Huang, Y., Wang, J., 2021. Toxicological effects of microplastics and phenanthrene to zebrafish (Danio rerio). Sci. Total Environ. 757, 143730 https://doi.org/10.1016/j.scitotenv.2020.143730.

Zanuzzo, F.S., Sabioni, R.E., Marzocchi-Machado, C.M., Urbinati, E.C., 2019. Modulation of stress and innate immune response by corticosteroids in pacu (Piaractus mesopotamicus). Comp. Biochem. Physiol. A Mol. Integr. Physiol. 231, 39–48. https://doi.org/10.1016/j.cbpa.2019.01.019.

Zhang, P., 2012. Analysis of mouse liver glycogen content. Bio-Protocol 2 (10), 1–4. https://doi.org/10.21769/BioProtoc.186.

Zhang, C., Wang, J., Zhou, A., Ye, Q., Feng, Y., Wang, Z., Zou, J., 2021. Species-specific effect of microplastics on fish embryos and observation of toxicity kinetics in larvae. J. Hazard. Mater. 403, 123948.

Zhou, W., Chen, M.H., Shi, W., 2015. Influence of phthalates on glucose homeostasis and atherosclerosis in hyperlipidemic mice. BMC Endocr. Disord. 15, 13. https://doi.org/10.1186/s12902-015-0015-4.

Zuccarello, P., Ferrante, M., Cristaldi, A., Copat, C., Grasso, A., Sangregorio, D., Oliveri Conti, G., 2019. Exposure to microplastics (<10 μm) associated to plastic bottles mineral water consumption: the first quantitative study. Water Res. 157, 365–371. https://doi.org/10.1016/j.watres.2019.03.091.